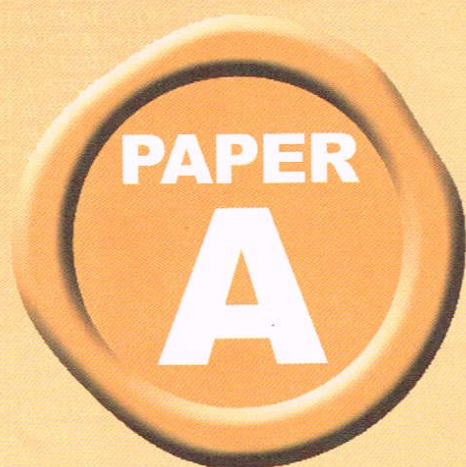




ICAS

INTERNATIONAL COMPETITIONS AND
ASSESSMENTS FOR SCHOOLS
MATHEMATICS
2008

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

**PLEASE SEE BACK COVER FOR A LIST
OF THE YEAR LEVELS THAT SHOULD**

SIT THIS PAPER

1.  Gold
Silver
Bronze

How many silver stars are there?

- (A) 2
(B) 4
(C) 5
(D) 11

2. This flower has 6 petals.



How many petals does this flower have?



- (A) 5
(B) 6
(C) 7
(D) 8

3. Three of these pictures show a footprint made by Lisa's left foot.

Which picture shows a footprint made by Lisa's right foot?



(A)

(B)



(C)

(D)

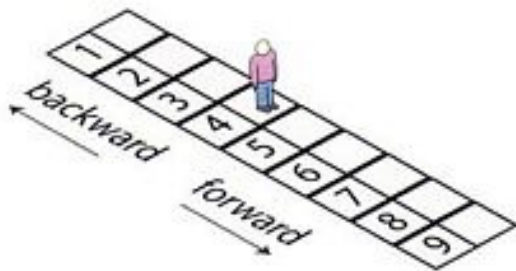
4. This is a picture of a house.



How many four-sided shapes are there in the picture?

- (A) 2
(B) 3
(C) 4
(D) 5

5. Bill is on square 4 in the picture.



He moves 5 squares forward and 3 squares backward and stops.

On which square does he stop?

- (A) 1
- (B) 2
- (C) 6
- (D) 9

6. $12 + 3 =$?

- (A) 14
- (B) 15
- (C) 16
- (D) 17

7. Joe had these pencils.

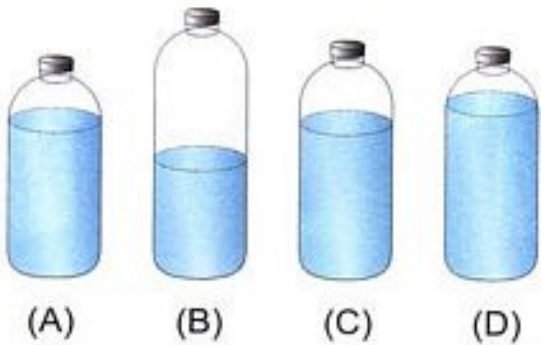


Joe lost two of the pencils.

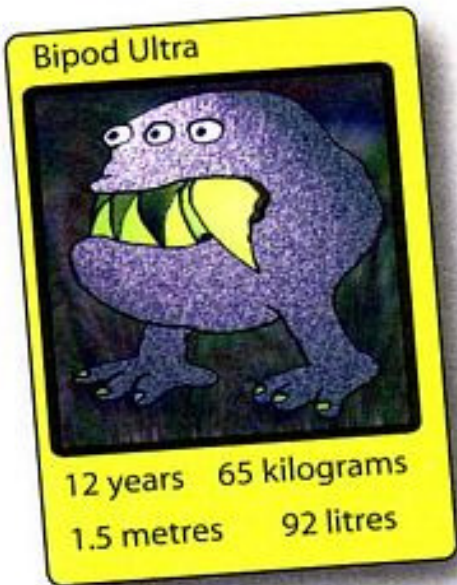
How many pencils are left?

- (A) 5
- (B) 6
- (C) 8
- (D) 9

8. The picture shows four bottles of water.
Which bottle has the most water in it?



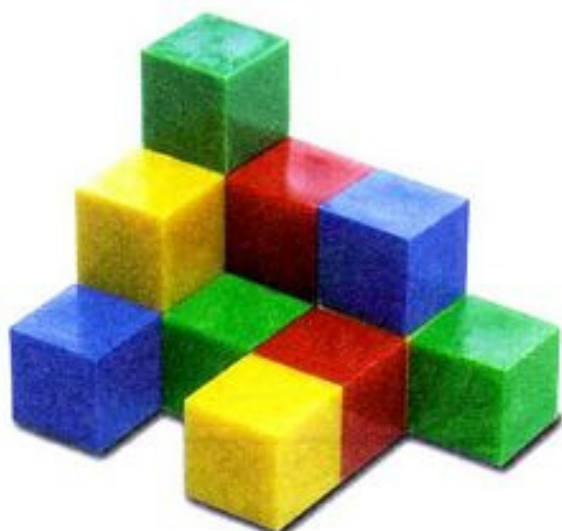
9. Finn collects monster cards like this one.



What is the mass of the monster?

- (A) 12 years
- (B) 65 kilograms
- (C) 1.5 metres
- (D) 92 litres

10. Meera made this shape out of cubes.



The length of the shape is 4 cubes.

What is the height of the shape?

- (A) 1 cube (B) 2 cubes
(C) 3 cubes (D) 4 cubes

11. Brian picks one of these counters without looking.



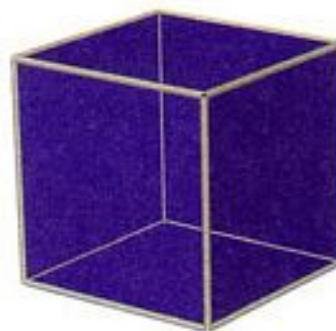
Which kind of counter is he **least** likely to pick?

- (A) (B) (C) (D)

12. $16 + 27 = ?$

- (A) 33
(B) 42
(C) 43
(D) 88

13. A cube has 6 faces.



Which solid has 8 faces?



(A)



(B)



(C)



(D)

14. What time is shown on this clock?



- (A) one past six
(B) six past one
(C) half past one
(D) six to two

15. What is the next shape in this pattern?



(A)



(B)



(C)



(D)

16. Ellen is thinking of a number that is bigger than 28 but smaller than 41.

Which of these numbers could be Ellen's number?

- (A) 13
(B) 25
(C) 39
(D) 45

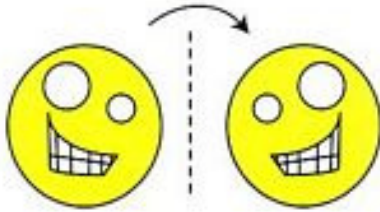
- 17.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

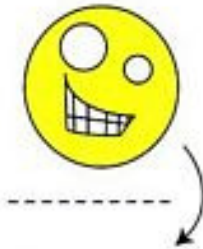
What fraction of this number square is shaded?

- (A) one-half
(B) one-quarter
(C) one-fifth
(D) one-tenth

18. The picture shows how Priya flipped a shape.



Priya flipped the same shape in a different way.



What will the shape look like after it has been flipped?



(A)



(B)



(C)



(D)

19. Mike uses this symbol $\blacktriangle$ for the number 23.

$$10 \times \blacktriangle + 17 = \text{?}$$

What number is ? ?

- (A) 27
- (B) 170
- (C) 247
- (D) 400

20. Jess cut an apple into four pieces.



Which one of these shows one whole apple cut into four pieces?



(A)



(B)



(C)



(D)

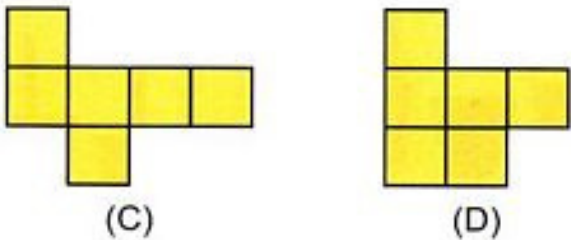
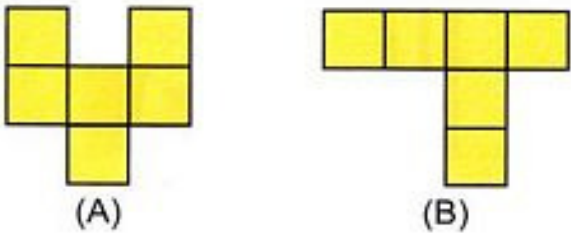
21. Here is one bunch of cherries.



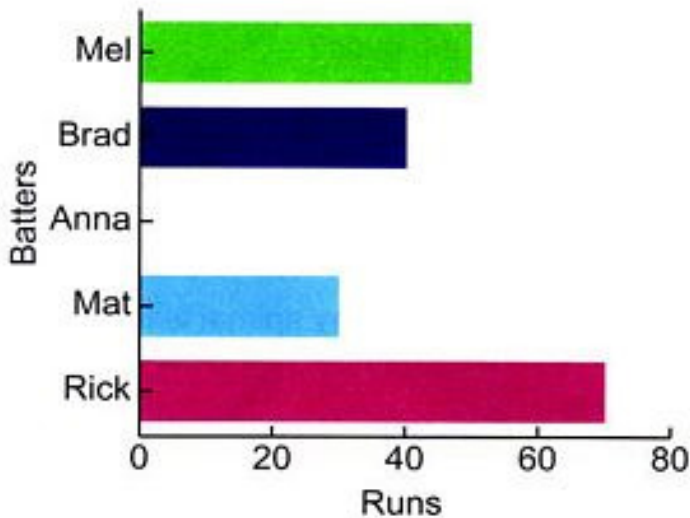
How many cherries in total are in 12 bunches like this one?

- (A) 3
- (B) 12
- (C) 24
- (D) 36

22. Which of the figures below can be folded along the lines to make a cube?



23. The graph shows the runs of some people playing cricket. Anna's runs are not shown. In total the five people scored 225 runs.



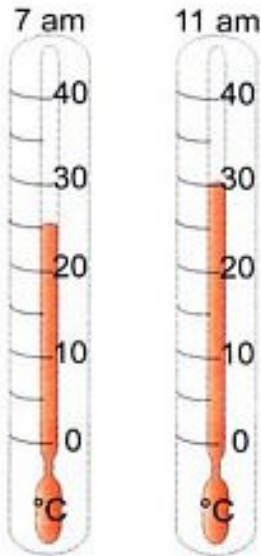
How many runs did Anna score?

- (A) 55
- (B) 45
- (C) 35
- (D) 25

24. Alice had 50 berries. She ate 17 berries.
How many berries did she have left?

- (A) 33
- (B) 37
- (C) 43
- (D) 47

25. Emma checked the temperature in her garden at 7 am and at 11 am.



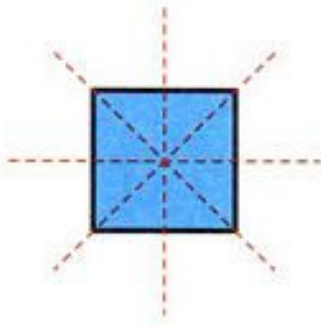
By how much had the temperature increased?

- (A) 1 °C
- (B) 5 °C
- (C) 10 °C
- (D) 30 °C

26. Which of these is the longest?

- (A) 1 month
- (B) 6 weeks
- (C) 27 days
- (D) 48 hours

27. A square has four lines of symmetry.



Which of these shapes has the most lines of symmetry?



(A)



(B)



(C)



(D)

28. Amy wrote down a number sentence, but one digit was rubbed out.

$$176 + 2\text{ }4 = 430$$

What is the missing digit?

- (A) 3
- (B) 4
- (C) 5
- (D) 6

29. Alice has 3 identical straws and 3 different cups to put them in.



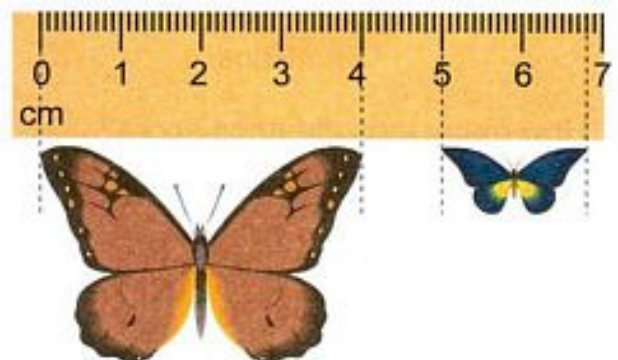
Here are four different ways she can put all the straws into the cups.



How many different ways can she put the straws into the cups?

- (A) 5
- (B) 6
- (C) 8
- (D) 10

30. The wingspan of any animal is the distance between the tips of its largest wings.



What is the **difference** in wingspan between these two butterflies?

- (A) 1.0
- (B) 2.2
- (C) 2.8
- (D) 5.0

31. Which of these is the same as 1200?

- (A) 12 millions
- (B) 12 thousands
- (C) 12 hundreds
- (D) 12 tens

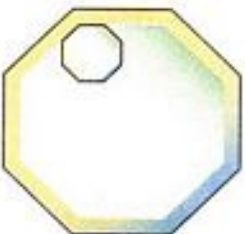
32. What will this shape look like after a half turn?



(A)



(B)



(C)



(D)

33. A class of 11 students has a packet of 44 jelly beans and another packet of 66 jelly beans.

The class puts all the jelly beans together and then shares them out. Every student gets the same number of jelly beans.

How many jelly beans does each student get?

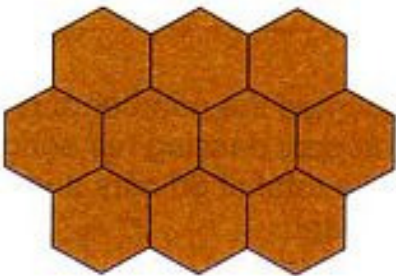
- | | |
|--------|---------|
| (A) 4 | (B) 6 |
| (C) 10 | (D) 110 |

34. In a shop, 3 bottles of orange juice cost the same as 2 bottles of mango juice.

How many bottles of orange juice cost the same as 6 bottles of mango juice?

- | | |
|-------|-------|
| (A) 3 | (B) 4 |
| (C) 8 | (D) 9 |

35. How many hexagons are touching **only** 3 other hexagons?



- (A) 4
- (B) 6
- (C) 7
- (D) 8

36. Here is a number sentence.

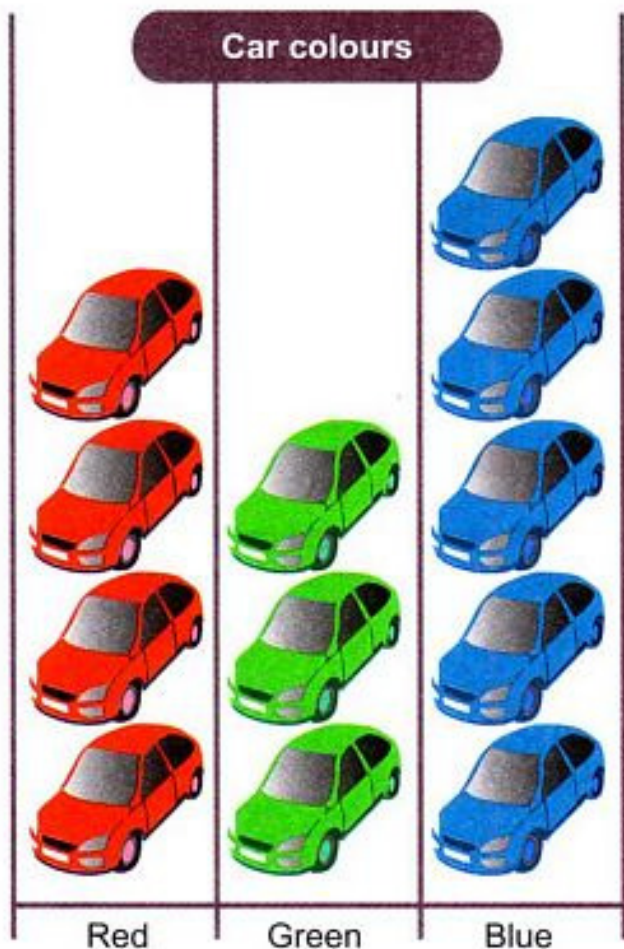
$$\frac{2}{\boxed{?}} \text{ is greater than } \frac{\boxed{?}}{10}$$

Which number could go in both boxes to make the number sentence true?

- (A) 4
- (B) 5
- (C) 6
- (D) 7

37. Peter has a pet bird. Every day he gives his bird 20 g of seed.
- On Thursday Peter opened a 750 g packet of seed for the first time and gave his bird 20 g of seed.
- What day of the week will he need to open a new packet?
- (A) Thursday
(B) Friday
(C) Saturday
(D) Sunday

38. Jane counted the number of red, green and blue cars passing her school. She counted a total of 60 cars and drew this picture graph.



How many green cars did she count?

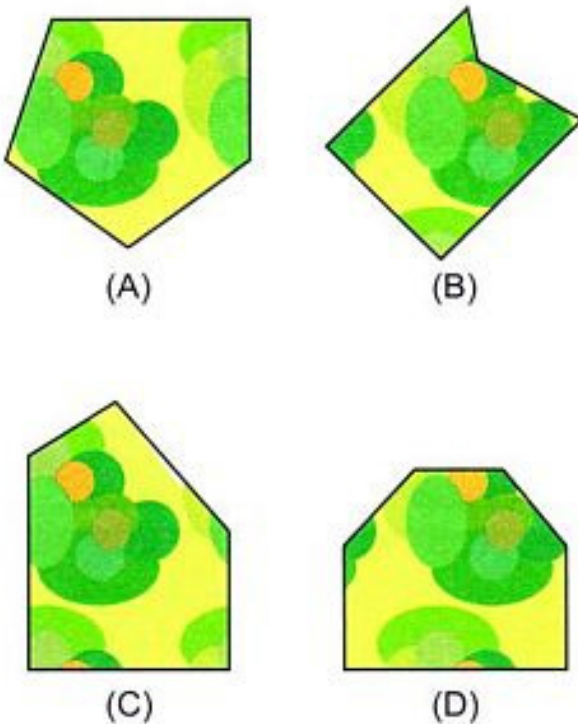
- (A) 3
(B) 5
(C) 12
(D) 15

39. A cat and a dog together have a mass of 42 kilograms.
- The dog's mass is 30 kilograms more than the cat's mass.
- What is the mass of the cat?
- (A) 6 kg
(B) 12 kg
(C) 24 kg
(D) 36 kg

40. Bill cut a shape out of paper.

His shape has:
5 sides, 2 right angles, and
2 sides the same length.

Which shape could be Bill's shape?



This page may be used for working.

ACKNOWLEDGMENT

Copyright in this booklet is owned by Educational Assessment Australia, UNSW Global Pty Limited unless otherwise indicated. Every effort has been made to trace and acknowledge copyright for materials used. Educational Assessment Australia apologises for any accidental infringement and welcomes information to redress the situation.

**THE FOLLOWING YEAR LEVELS
SHOULD SIT THIS PAPER:**

AUSTRALIA: Year 3
BRUNEI: Primary 3
INDONESIA: Year 4
MALAYSIA: Standard 3
NEW ZEALAND: Year 4
PACIFIC: Year 3
SINGAPORE: Primary 2
SOUTH AFRICA: Grade 3

**PAPER
A**

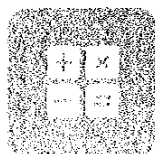




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

ICAS

International Competitions
and Assessments for Schools



2008 ICAS Mathematics Answer Keys

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
1	B	C	A	A	C	B	A	A
2	D	A	D	D	A	C	D	B
3	A	D	C	C	C	A	B	C
4	D	C	D	D	D	D	C	D
5	C	A	C	D	C	D	D	D
6	B	C	B	C	D	A	C	A
7	A	B	D	A	B	C	B	B
8	D	D	C	B	D	B	C	B
9	B	B	D	B	B	C	D	A
10	C	B	C	A	D	C	A	B
11	D	B	A	D	C	B	B	B
12	C	D	D	A	D	D	A	A
13	A	C	A	C	D	A	B	D
14	C	C	C	D	B	C	D	A
15	A	A	B	C	A	A	B	D
16	C	A	D	C	C	A	D	D
17	B	D	B	A	B	D	A	B
18	D	B	A	B	C	B	C	C
19	C	C	D	D	B	C	C	D
20	D	C	B	B	D	A	D	D
21	D	D	B	C	C	B	C	C
22	C	C	C	C	A	D	B	B
23	C	A	A	A	C	B	D	A
24	A	B	C	B	C	D	B	B
25	B	B	D	A	A	D	D	C
26	B	D	B	C	B	A	B	A
27	D	D	D	D	A	B	C	A
28	C	A	B	D	B	B	A	B
29	D	B	A	B	A	A	B	B
30	B	D	C	C	C	B	C	D
31	C	C	C	C	B	C	B	C
32	A	D	D	D	D	D	A	A
33	C	C	B	B	A	B	A	D
34	D	C	C	A	A	C	B	C
35	B	D	A	B	B	A	D	C
36	A	C	B	968	39	56	24	162
37	C	B	C	260	56	185	63	123
38	D	D	A	12	799	400	57	135
39	A	B	C	105	105	32	95	312
40	B	B	B	16	648	320	62	47

Contact

T: +61 (2) 8344 1016
E: info@eaa.unsw.edu.au
W: www.eaa.unsw.edu.au

© 2010 Educational Assessment Australia. EAA is an education group of UNSW Global Pty Limited, a not-for-profit provider of education, training and consulting services and a wholly owned enterprise of the University of New South Wales. ABN 62 086 418 582.